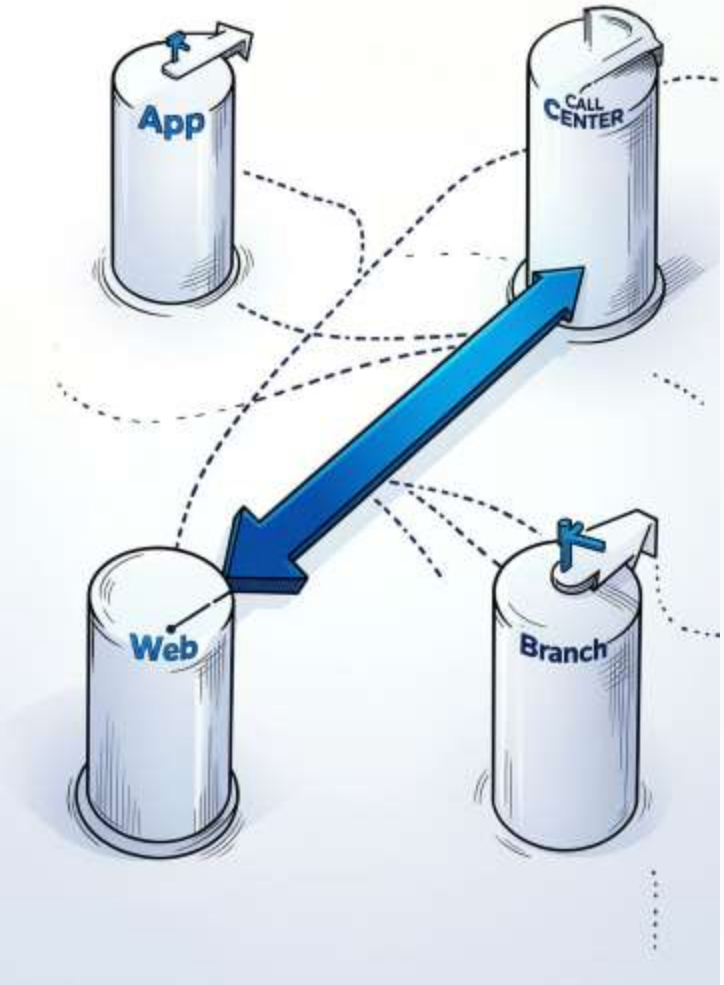


JourneySync

Real-Time Cross-Channel Identity Resolution & Customer Journey

From fragmented interactions to one continuous customer story—JourneySync unifies signals across App, Web, Call Center, Branch, and messaging channels into a single, real-time timeline so front-line teams act with context and confidence.

Tagline: One platform. One view. One customer story.

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The Cost of Broken Customer Visibility

A card member spends 20 minutes completing a credit-limit request, abandons it, and calls support an hour later. The agent has zero context. The customer repeats everything — frustration spikes and the opportunity is lost. This is not an anecdote; it's the daily reality for banks that can't see journeys.

What Banks Can't Do

- See the full customer journey
- Detect digital drop-off moments
- Intervene before churn

Why It Hurts

- Agents fly blind — longer calls, poorer outcomes
- UX failures go unfixed — conversion leaks persist
- Reactive operations — higher retention cost

Bottom line: Current systems understand isolated interactions. They do not understand journeys.



This Is Not an Edge Case

4–6

Channels per Issue

Every customer touches 4–6 channels before resolving a single issue.

67%

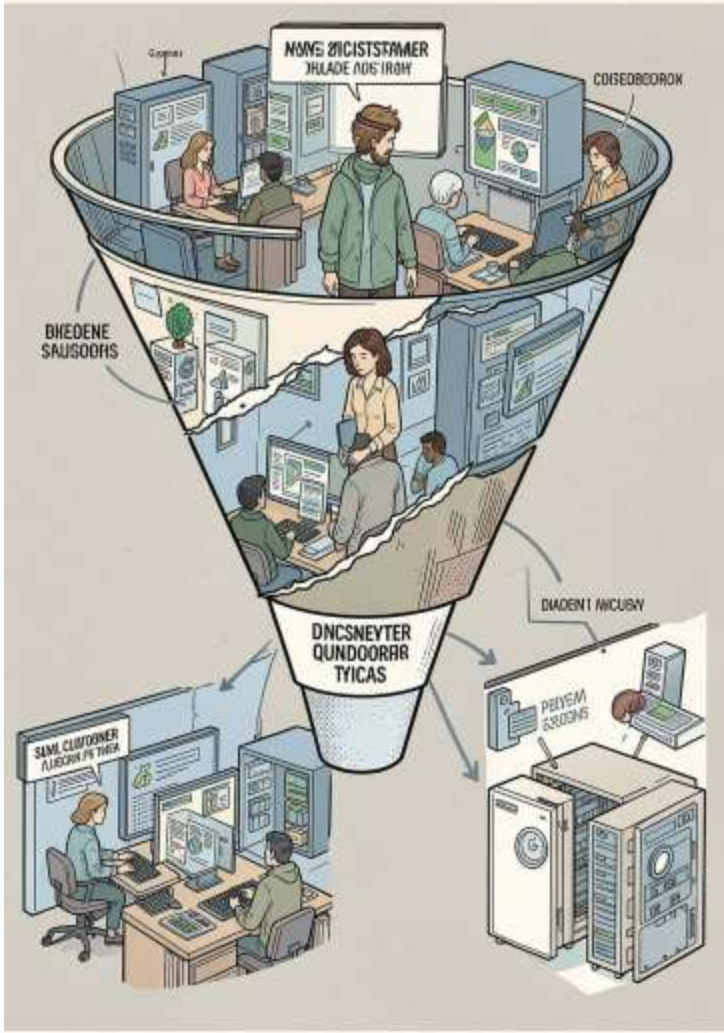
Form Abandonment

67% of customers who abandon a digital form never return.

$5–$15

Repeat Call Cost

Repeat calls cost banks $5–$15 per avoidable interaction.



Caption: The data exists — multiple records across systems — but the connection between them does not.

JourneySync — One Platform, One Customer View

Ingest

Capture raw events from Web, App, Call Center, Branch, Email, Chatbot, and third-party sources at millisecond scale.



Resolve

Deterministic + probabilistic matching merges anonymous sessions to customer profiles with confidence scoring.



Stitch

Assemble a single chronological timeline per customer for the entire lifecycle — discoverable, queryable, and auditable.



Act

AI detects stage, predicts next action, flags churn risk, and surfaces recommended operator actions in real time.



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How It Works

What it does

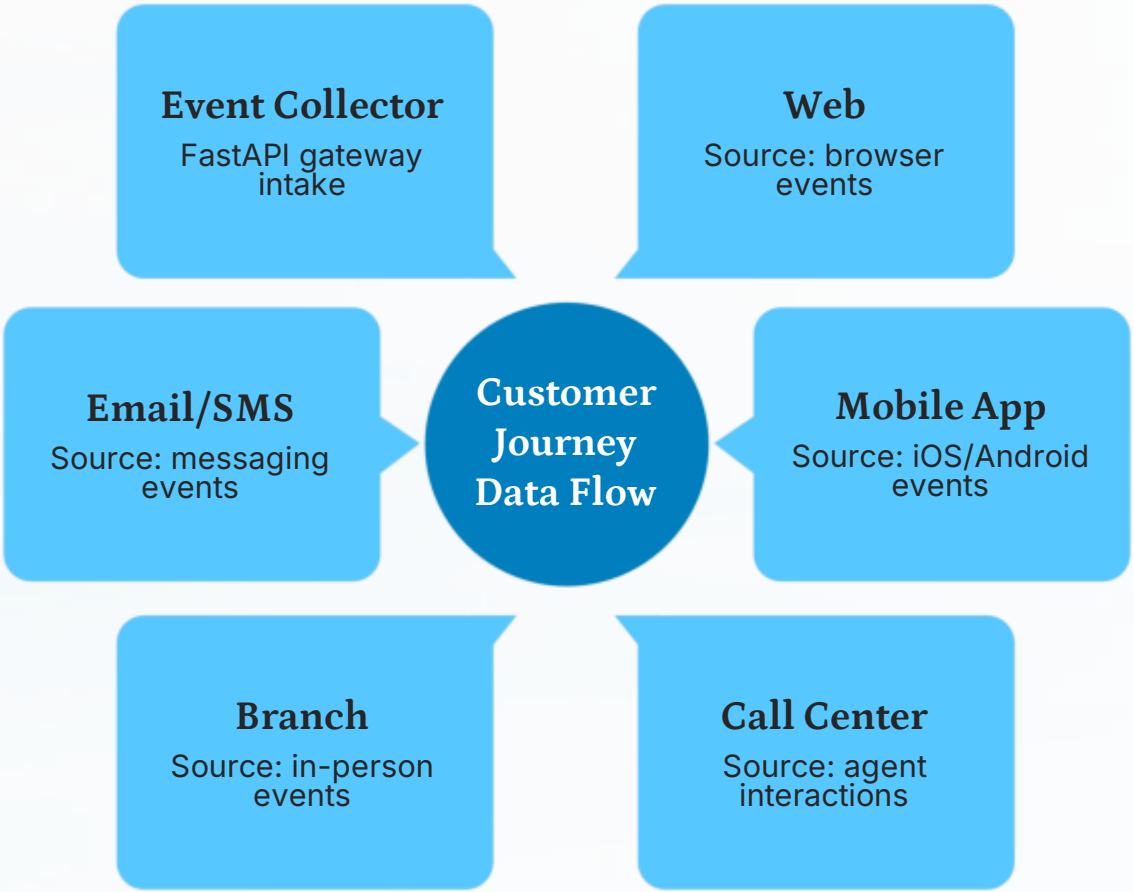
- Ingests raw events across channels and normalizes them to a unified schema.
- Resolves anonymous sessions to known profiles with deterministic + probabilistic matching.
- Stitches events into a chronological timeline per customer.
- Applies AI to detect journey stage, predict next action, and flag churn risk.
- Surfaces context in a real-time operator dashboard for immediate intervention.

Value prop: We don't just store what happened — we understand the journey.



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System Architecture



Event-driven, stateless, horizontally scalable. Average end-to-end latency: 280ms.

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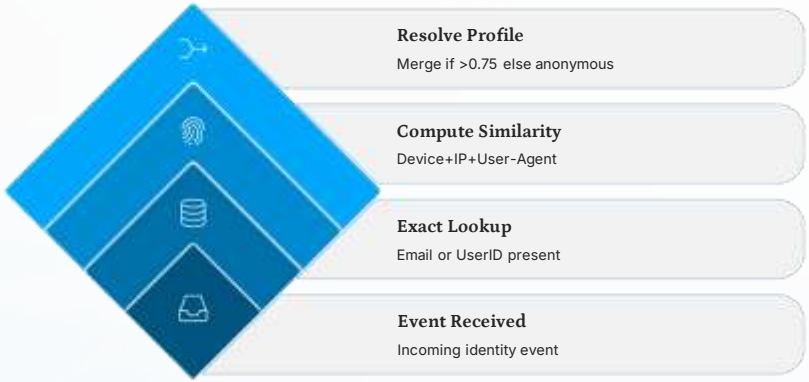
Identity Resolution Engine — The Brain

Identifier Mapping

Channel	Identifier Used
Website	Email / Session Cookie
Mobile App	Customer ID
Call Center	Phone Number
Branch	PAN / Customer Number
CRM	Loyalty Card Number

Mapping standardized across sources so every event carries canonical identifier fields for lookup and scoring.

Two-Stage Matching Flow

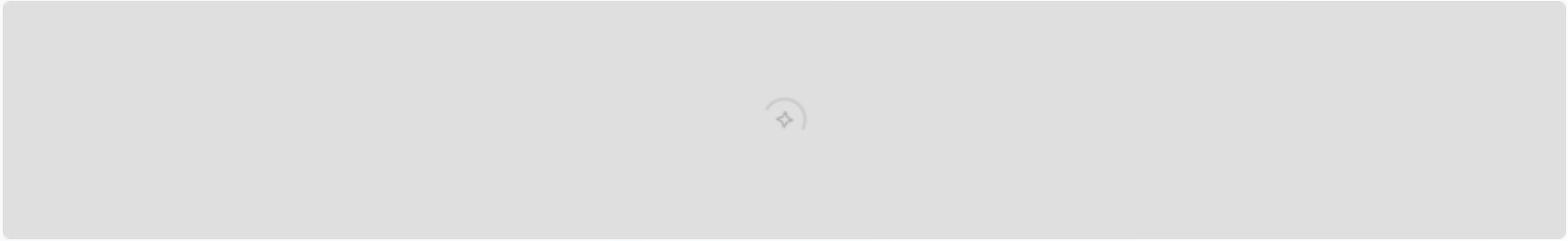


- **Deterministic Matching** — exact identifier lookup (instant, high confidence)
- **Probabilistic Matching** — weighted scoring on IP subnet, device fingerprint, temporal proximity
- **Confidence Score** — every stitched event carries a score; below 60% flagged for manual review

Result (synthetic tests): 98.4% precision · 96.7% recall on a 100-profile dataset.

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What a Stitched Journey Looks Like



This stitched timeline shows how separate interactions become one continuous customer journey across web, call center, email, and branch channels.

Before JourneySync

8 isolated records across 4 systems

After JourneySync

One continuous, chronological customer story

Current Stage detected by AI: **Application Under Review** — High priority for proactive outreach.

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AI Intelligence Layer — Beyond Aggregation



Intent Detection

What is the customer trying to accomplish right now?



Sentiment Analysis

Is frustration rising? Detect tone and urgency from text/voice transcripts.



Journey Stage Classification

Awareness → Consideration → Application → Onboarding mapped in real time.



Next Action Prediction

Probability of a return, call, or branch visit in the next 2 hours.



Recommendation Engine

Ranked operator actions (call, message, offer) to reduce churn or accelerate conversion.

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AI Models & Implementation

- XGBoost / LightGBM — Next action prediction (phase 1, deterministic features)
- Sentence Transformers / BERT — Intent & sentiment from transcripts and messages
- Rule-based engines → ML anomaly detection (phase 2) for fringe cases

Example output: Customer visited 'Cancel Account' twice + called twice in 24h → Churn Risk: HIGH → Recommended Action: Proactive retention call + $10 credit.

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Operator Dashboard — Designed for Instant Action

Mockup 1 — Customer Search

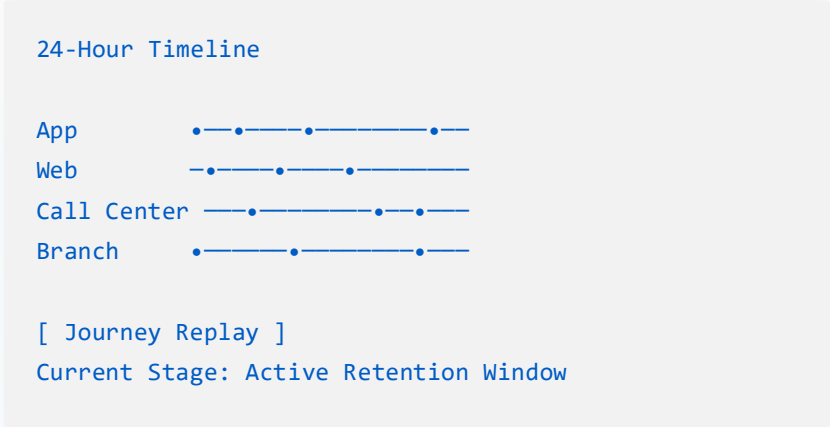
```
┌──────────────────────────────┐
│ Search by email, phone, ID... │
└──────────────────────────────┘


┌──────────────────────────────────────────────────────────┐
│ Customer   │ Last Seen │ Events Stitched │ Churn Score    │
├──────────────────────────────────────────────────────────┤
│ A. Smith   │ 10:42 AM  │ 24              │ 18             │
│ J. Lee     │ 09:15 AM  │ 31              │ 44             │
│ M. Patel   │ 08:03 AM  │ 19              │ 82  HIGH RISK  │
│ R. Diaz    │ 07:28 AM  │ 12              │ 27             │
└──────────────────────────────────────────────────────────┘
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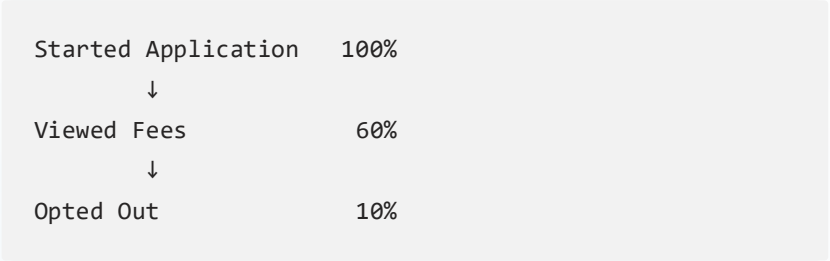
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Operator Dashboard — Timeline & Actions

Mockup 2 — Unified Timeline View



Mockup 3 — Drop-off Funnel + Alert Card



Action Required:

Customer visited Cancel page + called twice.

Recommend proactive retention call.

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Technology Stack — Modern, Enterprise-Grade

Layer	Technology	Why
Frontend	Next.js + Tailwind CSS	Fast, bank-grade UI
Visualizations	Recharts	Funnel & timeline charts
Backend	FastAPI (Python)	Async, high concurrency
Streaming	Redis Streams + Pub/Sub	Sub-500ms event handling
Identity DB	PostgreSQL	Relational identity graph
Timeline DB	MongoDB	Flexible nested documents, fast reads
Fuzzy Matching	fuzzywuzzy (Python)	Probabilistic string matching
Real-time Push	Server-Sent Events (SSE)	Live dashboard updates without polling
Mock Data	Python Faker	Synthetic test datasets for QA

Key metric: Average end-to-end pipeline latency: 280ms. Live dashboard updates stable under 500ms at target load.

Enterprise considerations: audit logging, encryption at rest/in transit, role-based access, and GDPR/PCI compliance built into data pipelines and storage.

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Built for Enterprise Scale

Stateless FastAPI Workers

Horizontally scalable behind load balancers, designed to handle 50,000+ events/second efficiently.

MongoDB Sharding

Distributes stitched timeline collections by customer ID, ensuring high availability and no single point of failure.

Redis Caching

Caches top 1,000 active customer profiles for instant dashboard loading and real-time responsiveness.

TTL Data Retention

Manages data lifecycle with 6-month hot storage for active timelines and automated cold archiving.

This illustrates JourneySync's linear scalability, maintaining a flat latency curve even with increasing concurrent users.

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Enterprise Security Built In

Control	Implementation
Authentication	JWT + OAuth upstream
Authorization	Role-Based Access Control (RBAC)
Data Privacy	PII masking on all analyst-facing views
Audit Trail	Immutable append-only event log
Transport	HTTPS / TLS everywhere
Confidence Flagging	Events below 60% confidence score flagged for human review
Data Governance	Customer data never leaves the bank's infrastructure

For financial institutions like AMEX, regulated data demands ironclad security. Unlike generic solutions, JourneySync is designed with these enterprise-grade controls from day one, making it deployable in the most sensitive environments.

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Expected Results

98.4%

Identity match

96.7%

Journey coverage

280ms

End-to-end Avg Latency

<500ms

Load Test (10K events/30s)

-20%

Repeat Calls (Expected impact)

+20%

App Completion (Expected impact)

Methodology note: Tested against a synthetic dataset of 100 profiles with known ground-truth mapping using Python Faker. Precision/recall calculated against ground truth.

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This Is Not Another CRM

Existing CRM Systems	JourneySync Platform
Stores isolated interactions	Reconstructs complete customer journeys
Static historical records	Real-time, continuous journey streams
Manual analysis required	AI-driven insights, actionable recommendations
Reactive — responds after escalation	Proactive — intervenes before escalation
Single-channel view	Unified cross-channel customer view
No confidence scoring	Every event carries a confidence score
No drop-off detection	Automatic funnel analysis & drop-off detection

Bottom line: CRMs are data repositories. JourneySync is an intelligence engine.

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Measurable ROI Across Every Stakeholder

1 Customer

- No more repeating information across channels
- Faster, context-aware support
- Proactive outreach before frustration peaks

2 Support Teams

- Full journey context before picking up the phone
- Automated churn risk alerts with recommended actions
- Reduced average handle time

3 Product Teams

- Pinpoint exact UX drop-off steps (e.g., 70% abandon at "Employment Details")
- Data-driven prioritization of fixes

4 Management

- Channel effectiveness analytics
- Customer lifetime value signals
- Operational efficiency metrics

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Future Roadmap

JourneySync is continuously evolving, with planned enhancements designed to deepen customer insights and expand operational capabilities.

Phase 1: Foundation

Journey Stitching, Identity Resolution, and the Analyst Dashboard are live now, forming the core platform.

Phase 2: Real-Time Intelligence

Integrate real-time Kafka streaming and ML-based anomaly detection within 1 month.

Phase 3: AI-Powered Support

Develop an LLM Agent for automated customer support context briefing within 2 months.

Phase 4: Enhanced Security

Implement predictive fraud detection leveraging journey anomalies in 2.5 months.

Phase 5: Ecosystem Integration

Enable enterprise CRM integration via robust REST APIs within 4 months.

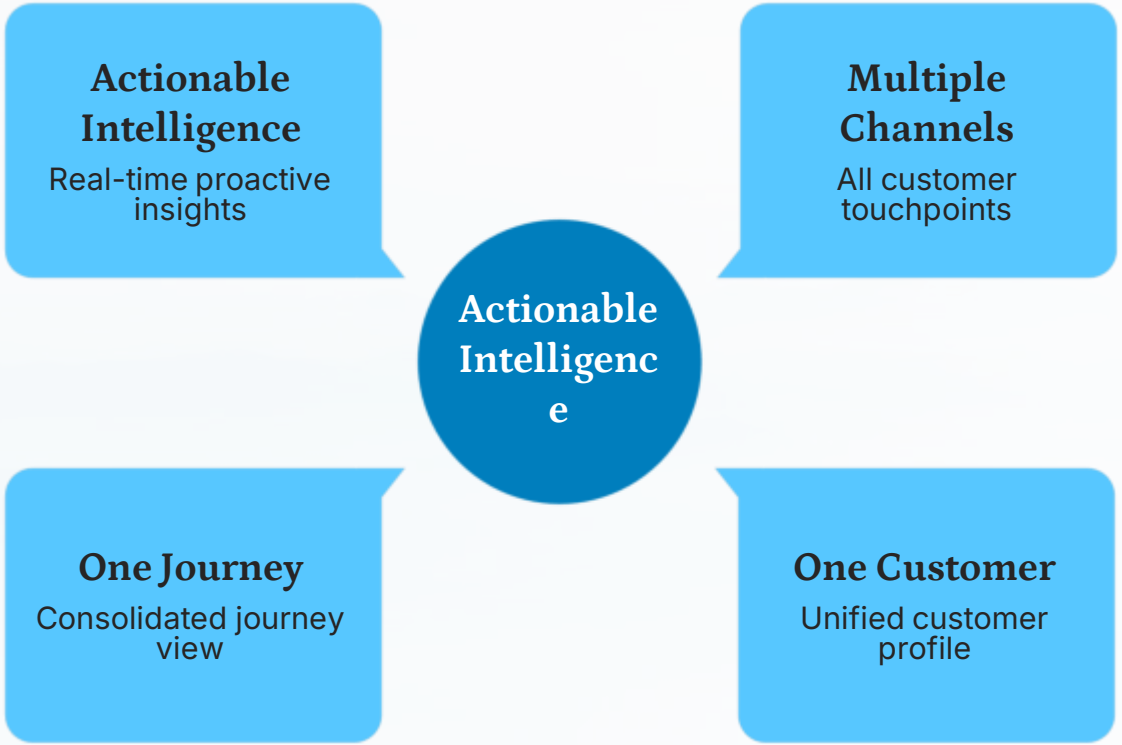
Phase 6: Advanced Simulation

Build a comprehensive Digital Twin of the customer for full behavioral simulation within 6 months.

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Closing Vision

Our vision is to transform fragmented customer interactions into a real-time journey that empowers financial institutions to deliver proactive, personalized, and seamless experiences across every touchpoint — before the customer ever has to ask twice.



Team: Cinephiles · **College:** IIT Mandi · **Members:** Vyom & Raj

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